Linear Algebra Problem Set #1

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Q1

1(a)

$$(AB)^{\mathrm{T}} = \left(\begin{bmatrix}2 & 0\\3 & 8\end{bmatrix}\begin{bmatrix}7 & 2\\6 & 3\end{bmatrix}\right)^{\mathrm{T}} = \left(\begin{bmatrix}14+0 & 4+0\\21+48 & 6+24\end{bmatrix}\right)^{\mathrm{T}}$$

$$= \left(\begin{bmatrix}14 & 4\\69 & 30\end{bmatrix}\right)^{\mathrm{T}} = \begin{bmatrix}14 & 69\\4 & 30\end{bmatrix}$$

$$B^{\mathrm{T}}A^{\mathrm{T}} = \begin{bmatrix}7 & 2\\6 & 3\end{bmatrix}^{\mathrm{T}}\begin{bmatrix}2 & 0\\3 & 8\end{bmatrix}^{\mathrm{T}} = \begin{bmatrix}7 & 6\\2 & 3\end{bmatrix}\begin{bmatrix}2 & 3\\0 & 8\end{bmatrix} = \begin{bmatrix}14+0 & 21+48\\4+0 & 6+24\end{bmatrix}$$

$$= \begin{bmatrix}14 & 69\\4 & 30\end{bmatrix}$$

1(b)

$$(AB)^{-1} = \begin{bmatrix}14 & 4\\69 & 30\end{bmatrix}^{-1} = \frac{1}{144}\begin{bmatrix}30 & -4\\-69 & 14\end{bmatrix}$$

$$B^{-1}A^{-1} = \frac{1}{9}\begin{bmatrix}3 & -2\\-6 & 7\end{bmatrix}\cdot\frac{1}{16}\begin{bmatrix}8 & 0\\-3 & 2\end{bmatrix} = \frac{1}{144}\begin{bmatrix}30 & -4\\-69 & 14\end{bmatrix}$$

1(c)

$$(A^{\mathrm{T}})^{-1} = \left(\begin{bmatrix}2 & 0\\3 & 8\end{bmatrix}^{\mathrm{T}}\right)^{-1} = \left(\begin{bmatrix}2 & 3\\0 & 8\end{bmatrix}\right)^{-1} = \frac{1}{16}\begin{bmatrix}8 & -3\\0 & 2\end{bmatrix}$$

$$(A^{-1})^{\mathrm{T}} = \left(\begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix}^{-1}\right)^{\mathrm{T}} = \left(\frac{1}{16}\begin{bmatrix} 8 & 0 \\ -3 & 2 \end{bmatrix}\right)^{\mathrm{T}} = \frac{1}{16}\begin{bmatrix} 8 & -3 \\ 0 & 2 \end{bmatrix}$$

Q2

This is not the case, in general. Consider the counterexample

$$\det(\mathcal{I}_n) + \det(\mathcal{I}_n) = 2 \neq 2^n = \det(\mathcal{I}_n + \mathcal{I}_n)$$

Q3

3(a)

The Hessian is

$$\begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$$

which has principal minors -2 and 3. Therefore it's negative definite and therefore it's strictly concave.

3(b)

The Hessian is

$$\begin{bmatrix} 6 & -2 & 0 \\ -2 & 4 & 2 \\ 0 & 2 & 2 \end{bmatrix}$$

It's easy enough to see that its first two principal minors are 6 and 20. The determinant is 16. Therefore, the matrix is positive definite and the function is strictly convex.

Q4

Symmetry

By the identities

$$\begin{aligned}
\mathcal{M}_{\mathbf{X}} &= \mathcal{I}_n - \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} \\
\mathcal{M}_{\mathbf{X}} - \mathcal{M}_{\mathbf{X}}^{\mathrm{T}} &= \mathcal{I}_n - \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} - (\mathcal{I}_n - \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}})^{\mathrm{T}} \\
&= \mathcal{I}_n - \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} - \mathcal{I}_n^{\mathrm{T}} + (\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}})^{\mathrm{T}} && \text{by Theorem 2(a)} \\
&= \mathcal{I}_n - \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} - \mathcal{I}_n + (\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}})^{\mathrm{T}} && \text{by } \mathcal{I}_n^{\mathrm{T}} = \mathcal{I}_n \\
&= -\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} + (\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}})^{\mathrm{T}} && \text{by Definition 15} \\
&= -\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} + (\mathbf{X}^{\mathrm{T}})^{\mathrm{T}}((\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1})^{\mathrm{T}}\mathbf{X}^{\mathrm{T}} && \text{by Theorem 2(c)} \\
&= -\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} + \mathbf{X}((\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1})^{\mathrm{T}}\mathbf{X}^{\mathrm{T}} && \text{by } (A^{\mathrm{T}})^{\mathrm{T}} = A \\
&= -\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} + \mathbf{X}((\mathbf{X}^{\mathrm{T}}\mathbf{X})^{\mathrm{T}})^{-1}\mathbf{X}^{\mathrm{T}} && \text{by Theorem 5.3} \\
&= -\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} + \mathbf{X}((\mathbf{X}^{\mathrm{T}}(\mathbf{X}^{\mathrm{T}})^{\mathrm{T}})^{\mathrm{T}})^{-1}\mathbf{X}^{\mathrm{T}} \\
&= -\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} + \mathbf{X}((\mathbf{X}^{\mathrm{T}}\mathbf{X})^{\mathrm{T}})^{-1}\mathbf{X}^{\mathrm{T}} \\
&= \mathbf{0}_n
\end{aligned}$$

and the definition of symmetry, the matrix $\mathcal{M}_{\mathbf{X}}$ is symmetric.

Idempotency

To show that $\mathcal{M}_{\mathbf{X}}$ is idempotent, write it as a difference of two matrices,

$$\mathcal{M}_{\mathbf{X}} = \mathbf{A} - \mathbf{B}, \qquad \text{where } \mathbf{A} = \mathcal{I}_n \text{ and } \mathbf{B} = \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}.$$

Each of the four products appearing in $(\mathbf{A} - \mathbf{B})^2$ simplifies:

$$\begin{aligned}
\mathbf{A}^2 &= \mathcal{I}_n\mathcal{I}_n = \mathcal{I}_n = \mathbf{A} \\
\mathbf{AB} &= \mathcal{I}_n\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} = \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} = \mathbf{B} \\
\mathbf{BA} &= \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathcal{I}_n = \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} = \mathbf{B} \\
\mathbf{B}^2 &= \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\underbrace{\mathbf{X}^{\mathrm{T}}\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}}_{=\mathcal{I}}\mathbf{X}^{\mathrm{T}} = \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} = \mathbf{B}
\end{aligned}$$

Expanding the square and substituting these four results gives

$$\begin{aligned}
\mathcal{M}_{\mathbf{X}}^2 &= (\mathbf{A} - \mathbf{B})^2 \\
&= \mathbf{A}^2 - \mathbf{AB} - \mathbf{BA} + \mathbf{B}^2 \\
&= \mathbf{A} - \mathbf{B} - \mathbf{B} + \mathbf{B} \\
&= \mathbf{A} - \mathbf{B} \\
&= \mathcal{M}_{\mathbf{X}}
\end{aligned}$$

so $\mathcal{M}_{\mathbf{X}}$ is idempotent.

Q5

The characteristic polynomial is

$$\begin{aligned}(.8-\lambda)(.95-\lambda)-(.05)(.2) &= .76-1.75\lambda+\lambda^2-.01\\ &= \lambda^2-1.75\lambda+.75\\ &= \frac{1}{4}(4\lambda^2-7\lambda+3)\\ &= \frac{1}{4}(\lambda-1)(4\lambda-3)\end{aligned}$$

Therefore, the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = \frac{3}{4}$

For $\lambda_1 = 1$:

$$\begin{aligned}(C-\lambda_1 I)u &= 0\\ \begin{bmatrix}0.8-1 & 0.05\\ 0.2 & 0.95-1\end{bmatrix}\begin{bmatrix}u_1\\ u_2\end{bmatrix} &= 0\\ -0.2u_1+0.05u_2 &= 0\\ u_2 &= 4u_1\end{aligned}$$

Normalizing the vector, we get the eigenvector for $\lambda_1 = 1$ is: $u = \begin{bmatrix}\frac{1}{\sqrt{17}} & \frac{4}{\sqrt{17}}\end{bmatrix}$

For $\lambda_2 = \frac{3}{4} = 0.75$:

$$\begin{aligned}(C-\lambda_2 I)u &= 0\\ \begin{bmatrix}0.8-0.75 & 0.05\\ 0.2 & 0.95-0.75\end{bmatrix}\begin{bmatrix}u_1\\ u_2\end{bmatrix} &= 0\\ 0.05u_1+0.05u_2 &= 0\\ u_2 &= -u_1\end{aligned}$$

Normalizing the vector, we get the eigenvector for $\lambda_2 = 3/4$ is: $u = \begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}}\end{bmatrix}$

Q6

The sum given in (a) is $\mathbf{u}'(\frac{1}{2}\Sigma^{-1})\mathbf{u}$ and the one given in (b) is $\mathbf{u}'\Lambda\mathbf{u}$.

Q7

The identities given in (a) follow directly from the definition of matrix multiplication.

The identity in (b) results from applying the identities in (a) to the identity $(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}X^{\mathrm{T}}\mathbf{y} = (\frac{1}{n}\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\frac{1}{n}\mathbf{X}^{\mathrm{T}}\mathbf{y}$ which follows from the commutativity and associativity of scalar-matrix multiplication and the identity $(\frac{1}{n}\mathbf{X})^{-1} = n\mathbf{X}^{-1}$

which can be obtained by applying the commutativity and associativity of scalar-matrix multiplication to show that $(\frac{1}{n}\mathbf{X})^{-1}n\mathbf{X}^{-1} = \mathcal{I}$.

(c) We have:

$$\begin{aligned}
\hat{\beta} &= (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}Y \\
\hat{\beta} &= (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}(X\beta + u) \\
\hat{\beta} &= (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}X\beta + (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}u \\
\hat{\beta} &= I\beta + (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}u \\
\hat{\beta} - \beta &= (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}u \\
\sqrt{n}(\hat{\beta} - \beta) &= \sqrt{n}(\frac{1}{n}X^{\mathrm{T}}X)^{-1}(\frac{1}{n}X^{\mathrm{T}}u) \text{ - this is because } (\frac{1}{n})^{-1} = n
\end{aligned}$$

We already had that $(X^{\mathrm{T}}X)^{-1} = (\sum_{i=1}^{n} x_i x_i^{\mathrm{T}})^{-1}$, and we also have $X^{\mathrm{T}}u = \sum_{i=1}^{n} x_i u_i$. Therefore, the formula is proved.

(d) We have that $\hat{u} = \mathcal{M}_{\mathbf{X}}\ y = \mathcal{M}_{\mathbf{X}}\ (X\beta + u) = \mathcal{M}_{\mathbf{X}}X\beta + \mathcal{M}_{\mathbf{X}}u$.

First, we want to prove that $\mathcal{M}_{\mathbf{X}}X\beta = 0$.

$$\begin{aligned}
\mathcal{M}_{\mathbf{X}}X\beta &= [I - X(X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}]X\beta \\
\mathcal{M}_{\mathbf{X}}X\beta &= X\beta - X(X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}X\beta \\
\mathcal{M}_{\mathbf{X}}X\beta &= X\beta - XI\beta \\
\mathcal{M}_{\mathbf{X}}X\beta &= 0
\end{aligned}$$

Therefore, $\hat{u} = \mathcal{M}_{\mathbf{X}}X\beta + \mathcal{M}_{\mathbf{X}}u = \mathcal{M}_{\mathbf{X}}u$.

Hence, $\sum(\hat{u}_i)^2 = \hat{u}^{\mathrm{T}}\hat{u} = (\mathcal{M}_{\mathbf{X}}u)^{\mathrm{T}}(\mathcal{M}_{\mathbf{X}}u) = u^{\mathrm{T}}(\mathcal{M}_{\mathbf{X}})^{\mathrm{T}}\mathcal{M}_{\mathbf{X}}u = u^{\mathrm{T}}\mathcal{M}_{\mathbf{X}}u$.

The last step is from the fact that $\mathcal{M}_{\mathbf{X}}$ is symmetric, therefore, $(\mathcal{M}_{\mathbf{X}})^{\mathrm{T}} = \mathcal{M}_{\mathbf{X}}$. $\mathcal{M}_{\mathbf{X}}$ is also idempotent, so $\mathcal{M}_{\mathbf{X}}\mathcal{M}_{\mathbf{X}} = \mathcal{M}_{\mathbf{X}}$.

Q8

Two functions that satisfy the specifications are below:

```r
### This function solve a linear system of equations: Ax = b
# A: nxn square matrix; b: nx1 vector
# If A is singular, print out "It does not have a unique solution"
# If A is not singular, output the solution

MC_solve = function (A, b) {

  ### Calculate det_A which is the determinant of A
  #Note: round to 10 decimal place, because even if A is singular,
  #det(A) may be a very small number not equal to 0 (because of R's algorithm)
  det_A = round(det(A), 10)

  ### Check if det_A is equal to 0, if not, solve the equation
  if (det_A == 0){
    print("It does not have a unique solution")
  } else {
    solution = solve(A, b)
    solution
  }
}

### For problem (a)
A1 = matrix(c(1, -2, 1, 0, 2, -8, -4, 5, 9), nrow = 3, byrow = TRUE)
b1 = c(0, 8, -9)

# Call the function on A1, b1
MC_solve(A1, b1)
### Result: (29, 16, 3)

### For problem (b)
A2 = matrix(c(1, 4, 2, 2, 5, 1, 3, 6, 0), nrow = 3, byrow = TRUE)
b2 = c(1, -2, 3)

# Call the function on A2, b2
MC_solve(A2, b2)
### Result: print "It does not have a unique solution"
```